

Sequences (7.1)

Name _____

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write “2” in the blank. Solve that problem, find your answer and write “3” in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: 4

_____ Find the 3rd term of the sequence $a_n = 2n + 1$.

ANSWER: $n^2 + 1$

_____ Simplify $\frac{8!}{6!}$.

ANSWER: $2n - 1$

_____ Evaluate $5!$.

ANSWER: 12

_____ Evaluate $0!$.

ANSWER: 7

_____ Write an explicit formula for the n th term of $1, 3, 5, 7, \dots$

ANSWER: 18

_____ Evaluate $\sum_{k=1}^3 2k$.

ANSWER: 10

_____ Write an explicit formula for the n th term of $2, 5, 10, 17, \dots$

ANSWER: 1

_____ Find the 4th term of $a_n = (-1)^n n$.

ANSWER: 120

_____ Evaluate $\sum_{k=1}^4 k$.

ANSWER: 56

_____ For $a_1 = 2$ and $a_k = 3a_{k-1}$, find a_3 .